INDR 252 APPLIED STATISTICS
DEPARTMENT OF INDUSTRIAL ENGINEERING
KOÇ UNIVERSITY
SPRING 2025

SYLLABUS

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Lerzan Ormeci



- BS&MS in IE (ODTÜ)
- PhD in Operations (Case Western Reserve University)
- Post-doctorate at EURANDOM in the Netherlands (A research institute on random processes)
- At Koç University Industrial Engineering since 2001
- Research interests: Health care applications, stochastic models
- Personal interests: Books, cats, volunteering, women in STEM



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PLAN OF THE DAY

- Syllabus
- Sampling and statistics

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INDR 252 APPLIED STATISTICS

Teaching assistants: Fatmanur Akdeniz, Arash Gandomkar Ghalhar, Ece Kuyucu, Ulvi Rajabov

Textbook: Statistics for Engineers and Scientists, 5th Edition (ISBN-13: 978-1260547887), W. Navidi, McGraw-Hill Education

McGraw-Hill Education Connect: Textbook companion, https://connect.mheducation.com/connect/container/launch.htm

You need buy the book and access code to be able to accomplish the reading assignments etc.

Software to carry on the statistical analyses: Python

- Num Python:

https://numpy.org/doc/stable/reference/routines.statistics.html

- Sci Python: https://docs.scipy.org/doc/scipy/reference/stats.html

All information about the course: LearnHub

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COURSE DESCRIPTION

- Visual tools to understand the data: all kinds of charts
- How to summarize data: useful statistics such average, standard deviation, etc.
- How to characterize the main properties of processes using data: parameter estimation, confidence intervals, and hypothesis testing.
- How to characterize the underlying probabilistic features using data: Distribution fitting, goodness of fit tests. Independence tests and contingency tables.
- How to relate different variables: Simple linear regression and correlation analysis. Nonlinear and multiple regression, analysis of categorical data.

Industrial engineering applications in quality control and demand forecasting. Statistical software packages and computer implementations.

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UPON COMPLETION OF THIS COURSE, YOU SHOULD BE ABLE TO:

- Formulate a problem about a system or process as a statistical question
- Design an appropriate data set and collect the data
- Present the data in visual forms that help understanding potential problems
- Understand and carry on the required statistical analyses to answer the chosen question
- Note the shortcomings of the statistical analyses due to the underlying assumptions and data collection process
- Assess statistical inferences on the underlying problem.

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LEARNING EXPERIENCES

- Learning material: introducing the statistical techniques that IEs use
- Experiencing:
  - Letting the students experience being an IE through cases.
  - Labs and case studies are designed so that students actively analyze data using appropriate software.

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Tentative schedule of the course

Spring 2025	Tue	Thu	Lab Content
Week 1	Introduction	Random sampling and data description	Lab 1
Week 2	Random sampling and data description	Propogation of error	Lab 2 - Generating random variables-
Week 3	Propogation of error	Point estimation	Lab 3 - Analyzing Dataset
Week 4	Point estimation	Probability plots/CLT	Lab 4 - CI
Week 5	Interval estimators	Interval estimators	Sample Case
Week 6	Interval estimators	Interval estimators	Case Exam 1
Week 7	**Spring Break**		
Week 8	Interval estimators	Interval estimators	PS
Week 9	Tests of hypotheses	Tests of hypotheses	Lab 5 - HT
Week 10	Tests of hypotheses	Tests of hypotheses	Sample Case
Week 11	Tests of hypotheses	**1-May**	**1-May**
Week 12	Linear regression	Linear regression	Lab 6 - ChiSq Distribution Tests
Week 13	Linear regression	Linear regression	PS
Week 14	Multiple regression	Multiple regression	Lab 7 - Multiple Regression
Week 15	Multiple regression	Multiple regression	Case Exam 2

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ASSESSMENT METHODS

Method	Description	Weight %
Case Study 1	Case Exam	10.00
Case Study 2	Case Exam	15.00
Midterm	Exam	25.00
Final Exam	Comprehensive Final exam	30.00
Reading Assignments	Assignments from the book	5.00
Participation	Participation (in-class exercises)	5.00
Participation	Lab work (Lab assignments)	10.00

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LECTURES AND LABS

- **Lectures:** In-class participation: class exercises (5%)
- **Lab Sessions** (10%)**:** Thursdays at 17:30-18:40 in computer lab SNAB149
  - You will have access to a recording of these sessions, as well as all related files, on the previous Monday. All lab sessions, except for the first one, have a short assignment, which are due to the end of each lab session. During the lab sessions, we will be present in the lab to answer your questions and help you complete your assignments. If a lab assignment is submitted **k** days later than the due date, the assignment will be evaluated out of (100-15k) %, where k can be **at most 2**.
  - Problem sessions: problems on the course topics will be solved by the TAs (2 PSs)
  - Sample cases: all students should read and think about the case before the session; during the sessions random groups will work on one part of the case and then we will discuss the results altogether. You will be given a short test on Blackboard at the end of this session. A sample case report will be uploaded to Blackboard after the session.
  - Case exams: in two sessions.

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READING ASSIGNMENTS (5%):

- The corresponding book chapters are assigned for reading before/after the lectures, so that you can be prepared for the new subjects and digest the material covered in class.
- Once you read the related chapters, you will be presented a number of questions on the basic concepts. We will hold Q&A sessions in the beginning of each lecture on the latest reading assignment to discuss your questions and comments on these concepts.
- These assignments will help you to assess your level of understanding in the course.

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LEARNHUB AND CONNECT

- How to register connect: Course Content/Connect Student Registration.
- E-book: Course Content/McGraw-Hill Connect Library and Assessments/McGraw-Hill Connect Library.
- Reading assignments:
  - Aim to help you in following the course.
  - Some midterm questions will be similar to these exercises.

If you cannot afford buying the book, please contact me.

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EXAMS

- **Midterm (25%):** multiple choice questions
- **Two case exams (10% & 15%):** Group exam; outputs are Python file and report
- **Final exam (30%)**: A comprehensive exam with multiple choice questions
- **Make-up Exam Policies:** A make-up exam for case and a make-up exam for other tests will be given to those who have a valid excuse at the end of the semester. Both make-up exams will be comprehensive.

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ACADEMIC DISHONESTY

- Koç University expects all its students to perform course-related activities in accordance with the rules set forth in the Student Code of Conduct (http://vpaa.ku.edu.tr/academic/student-code-of-conduct). Actions considered as academic dishonesty at Koç University include but are not limited to cheating, plagiarism and collusion. Copying from others or providing answers or information, written or oral, to others is cheating. Copying from another student's paper or from another text without written acknowledgment is plagiarism. Unauthorized help from another person or having someone else write one's paper or assignment is collusion.
- Cheating, plagiarism and collusion are serious offenses resulting in an F grade and disciplinary action.

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OTHER ISSUES

- **Grades:** To be announced in LearnHub. All objections to grades should be made within 7 days of their announcement.

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